CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in building with LLMs, from multi-agent workflows and AI API integration to Python backend services in production, using AI coding tools such as Claude Code. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by a habit of measuring results and tracing problems to their root cause.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present | Muldestausee, Germany

- Designed and optimized a complex agentic system for knowledge extraction (LangGraph: entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel) over regulatory and technical documents, raising precision/recall from 33%/45% to 93%/90% against an SAP-system-derived ground truth.
- Developed Expert Interface, a full stack internal tool for graph curation (React front end, Python microservices back end) deployed on Azure, letting domain experts validate and approve knowledge-graph changes.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume data source, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Sole developer owning the full knowledge-extraction pipeline end-to-end: data ingestion, chunking, prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and working with SAP domain experts.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Worked as an AI-augmented engineer with current AI tools (Claude Code, custom skills, agentic coding workflows), iterating fast on code grounded in clean code principles.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, MCP, RAG, GraphRAG, React, FastAPI, Postgres, MongoDB, Neo4j, Vector Search, Docker, Kubernetes, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 | Leverkusen, Germany

- Worked with business stakeholders to understand their decision-making workflows, identified where agentic AI could help, then designed and deployed a LangGraph application that routes each request to vector database queries, general inference or summarization.
- Developed a React front end and a Python back end with FastAPI, using Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK, following clean code practices; version-controlled in GitLab with CI/CD.
- **Tech stack: Python, React, FastAPI, LangGraph, LangChain, Postman, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 | Leverkusen, Germany

- Automated evaluation of RAG generation and retrieval with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Built AWS cloud infrastructure with Lambda, SQS and DynamoDB for PrismaChat RAG evaluation using CDK.
- Benchmarked multilingual embedding models on over 1M chunks of Covestro's internal chemical data.

- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing and fine-tuning models for optimized retrieval.
- **Tech stack: Python, LangChain, Hugging Face Transformers, RAGAS, PytrecEval, PyTorch, Vector Databases, Azure AI Search, AWS (S3, Lambda, SQS, DynamoDB, CDK), Docker, Streamlit**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) for the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Matplotlib, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for time series analysis of EV load.
- Developed an LSTM that achieved 96% test accuracy with strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Growth Ops Assistant | TypeScript | Node.js | MCP | React | Postgres

- Internal AI tool built for a marketing and user-acquisition use case: a TypeScript (Node.js) MCP server with Zod-typed tools over synthetic campaign, offer and payout data in Postgres, retrieval over team playbooks, a React + TypeScript screen and a Slack bot.
- Provider-agnostic model layer so the model can be swapped by config, numbers served only by deterministic tools, and an eval set in GitHub Actions that gates prompt and model changes.

TECHNICAL SKILLSET

- **AI & Agents:** Agentic systems (LangGraph, multi-agent), AI APIs (LiteLLM, LangChain, Hugging Face Transformers), MCP, RAG, GraphRAG, prompting and prompt engineering, evaluation (RAGAS, PytrecEval)
- **Full Stack:** Python, FastAPI, Python microservices, REST APIs, React (beginner), Streamlit, Postman
- **Databases:** SQL, Postgres, MongoDB, Neo4j, Vector Databases (Qdrant, Pinecone, Azure AI Search)
- **Cloud & Tools:** Docker, Kubernetes, Azure, AWS (S3, EC2, Lambda, SQS, DynamoDB, CDK), Git, GitLab CI/CD, Claude Code (custom skills, agentic coding), Agile
- **ML:** Scikit-Learn, PyTorch, TensorFlow, XGBoost, time series (ARIMA, Prophet, LSTM)
- **Languages:** English (Business Fluent), German (B1), Hindi (Fluent), Telugu (Native)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering (Minor in Computer Science & Engineering) — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India